



Disaster or Opportunity: The Impact of the Pandemic on Graduate Employment

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Background & Literature Review

- COVID-19 pandemic impacted a variety of U.S. industries
- Significant growth in the technology, engineering, and medical industries
- Lockdowns restricted industries such as transportation, tourism, customer service and the creative arts

Goals & Hypotheses

- How does the pandemic affect the graduate salary?
- Do different fields of study experience varying impacts from the pandemic?

Dataset Overview

- 229188 rows, 174 columns

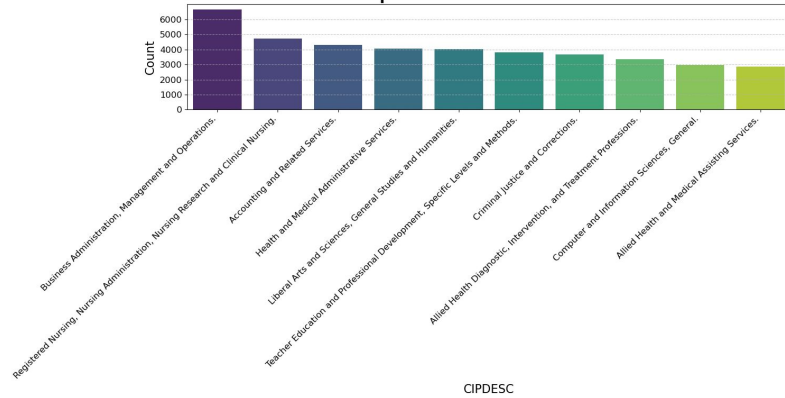
	UNITID	OPEID6	MAIN	CIPCODE	CREDLEV	IPEDSCOUNT1	IPEDSCOUNT2	DISTANCE
count	219079.000000	229188.000000	229188.000000	229188.000000	229188.000000	190264.000000	194383.000000	229188.000000
mean	202261.491435	6653.955150	0.955883	3321.287332	3.269547	27.869445	27.830011	1.080790
std	87098.757168	9089.097331	0.205355	1712.980303	1.914634	96.913120	100.235987	0.740192
min	100654.000000	1002.000000	0.000000	100.000000	1.000000	0.000000	0.000000	0.000000
25%	149231.000000	2029.000000	1.000000	1433.000000	2.000000	1.000000	1.000000	1.000000
50%	187532.000000	3151.000000	1.000000	4001.000000	3.000000	7.000000	7.000000	1.000000
75%	220978.000000	6661.000000	1.000000	5107.000000	5.000000	23.000000	23.000000	1.000000
max	497338.000000	43027.000000	1.000000	6127.000000	8.000000	9082.000000	9606.000000	3.000000

Dataset: Key Variables

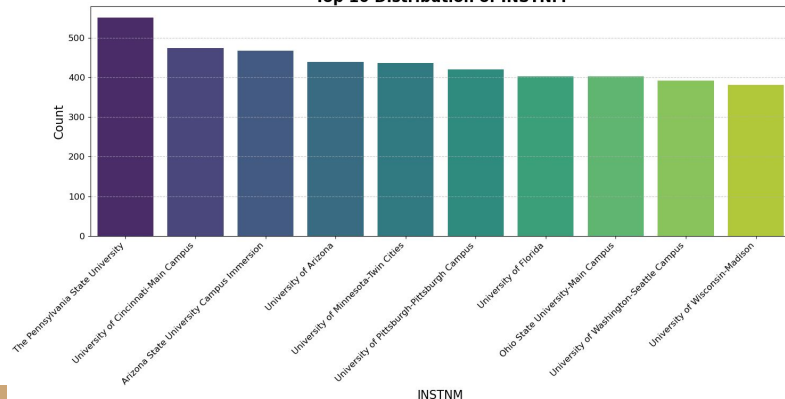
CONTROL	Working Sector (Public, Private Non-profit, Private For-profit, Foreign)
EARN_MDN_1YR/4YR/5YR	Median salary after 1 year, 4 years, 5 years
DISTANCE	Geographical distance
CIPDESC	Industry/Field of Work
CREDESC	Highest education attained (ex. Bachelors, masters, etc)
INSTNM	Institution where degree was earned
OPEID6	Office of Postsecondary Education Identifier (6 digits)
IPEDSCOUNT1/2	number of recognized postsecondary credentials conferred by field of study for two consecutive award years
UNITID	Integrated Postsecondary Education Data System ID Number
CIPCODE	Four-digit Classification of Instructional Programs Code

Variable Distributions

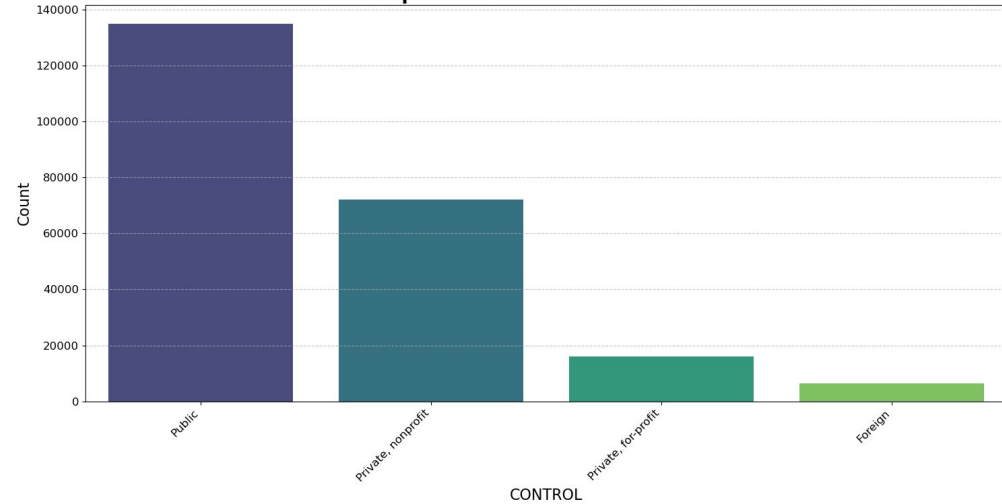
Top 10 Distribution of CIPDESC



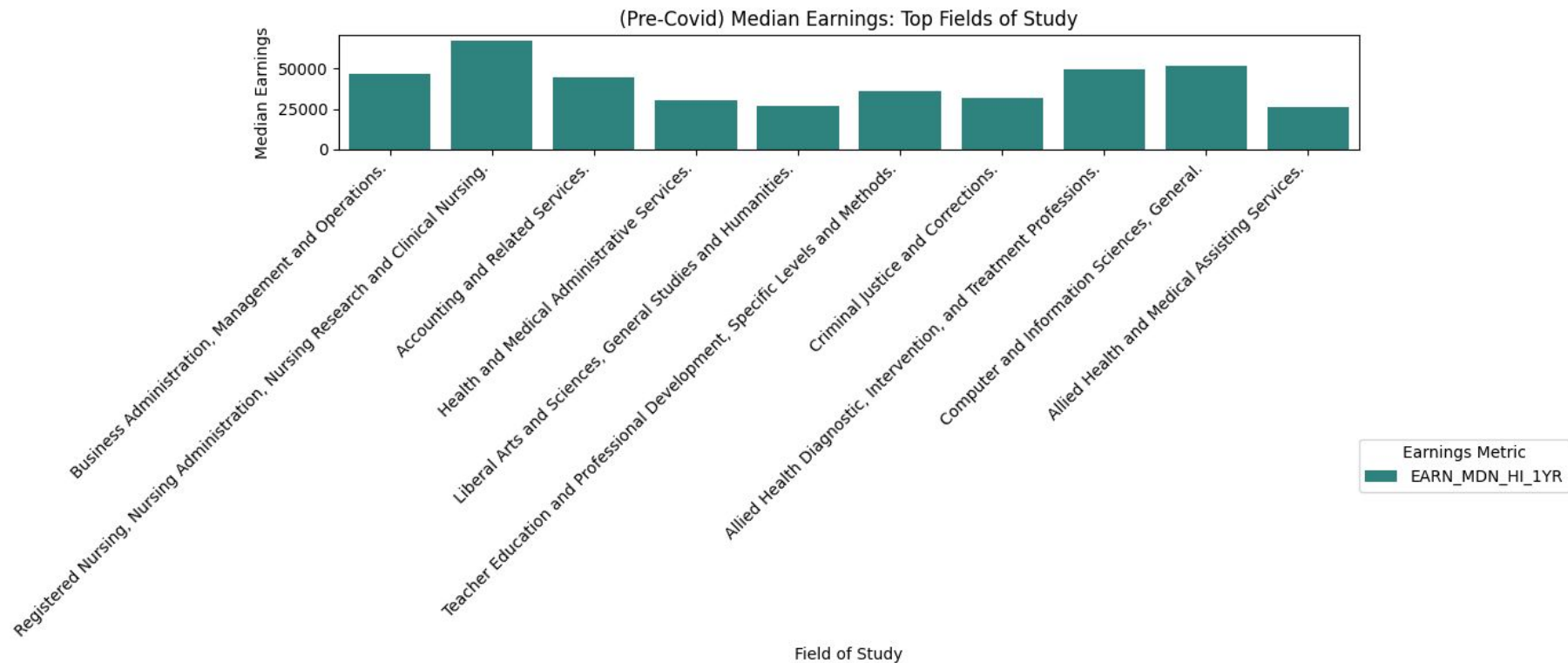
Top 10 Distribution of INSTNM



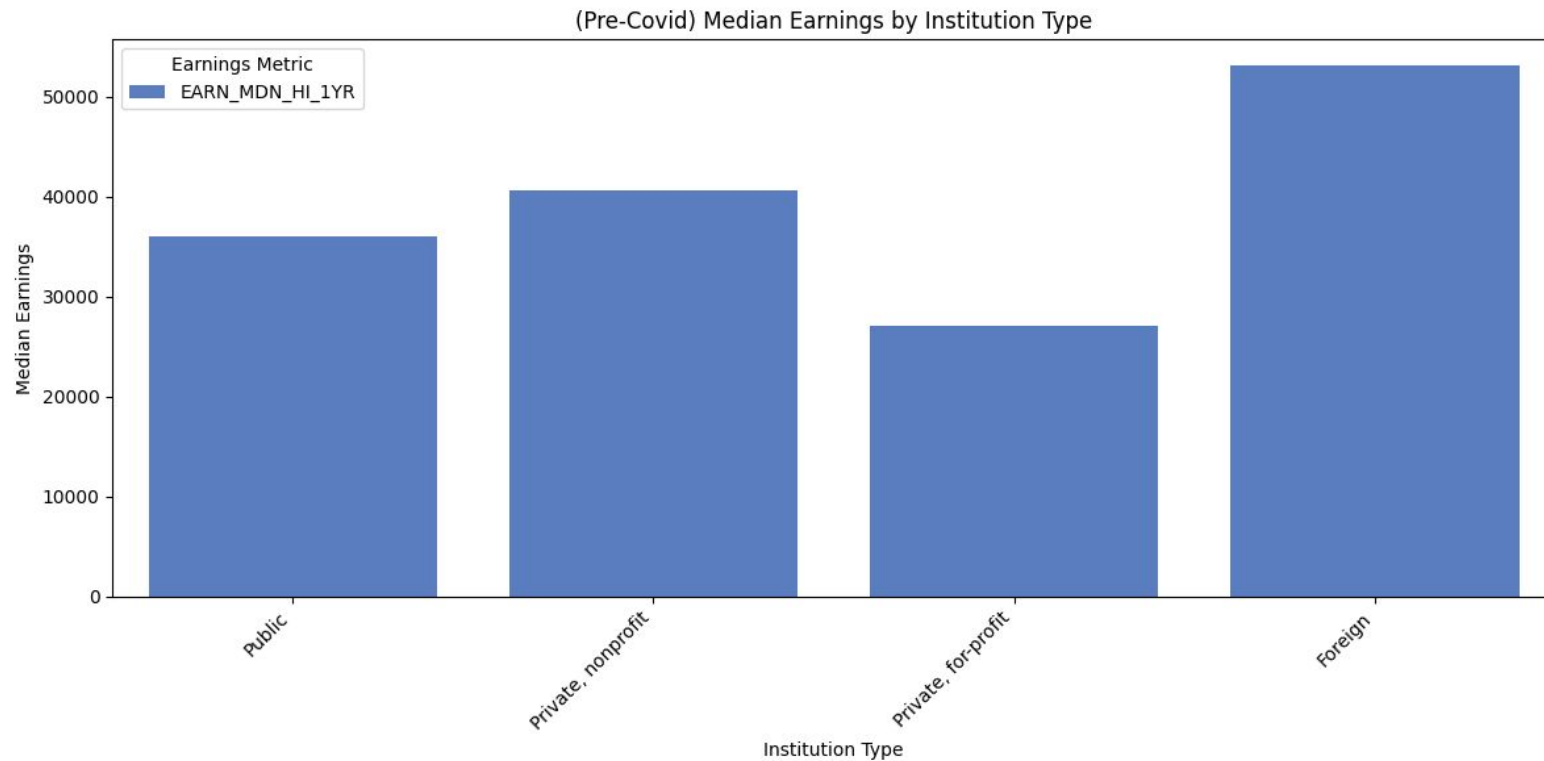
Top 10 Distribution of CONTROL



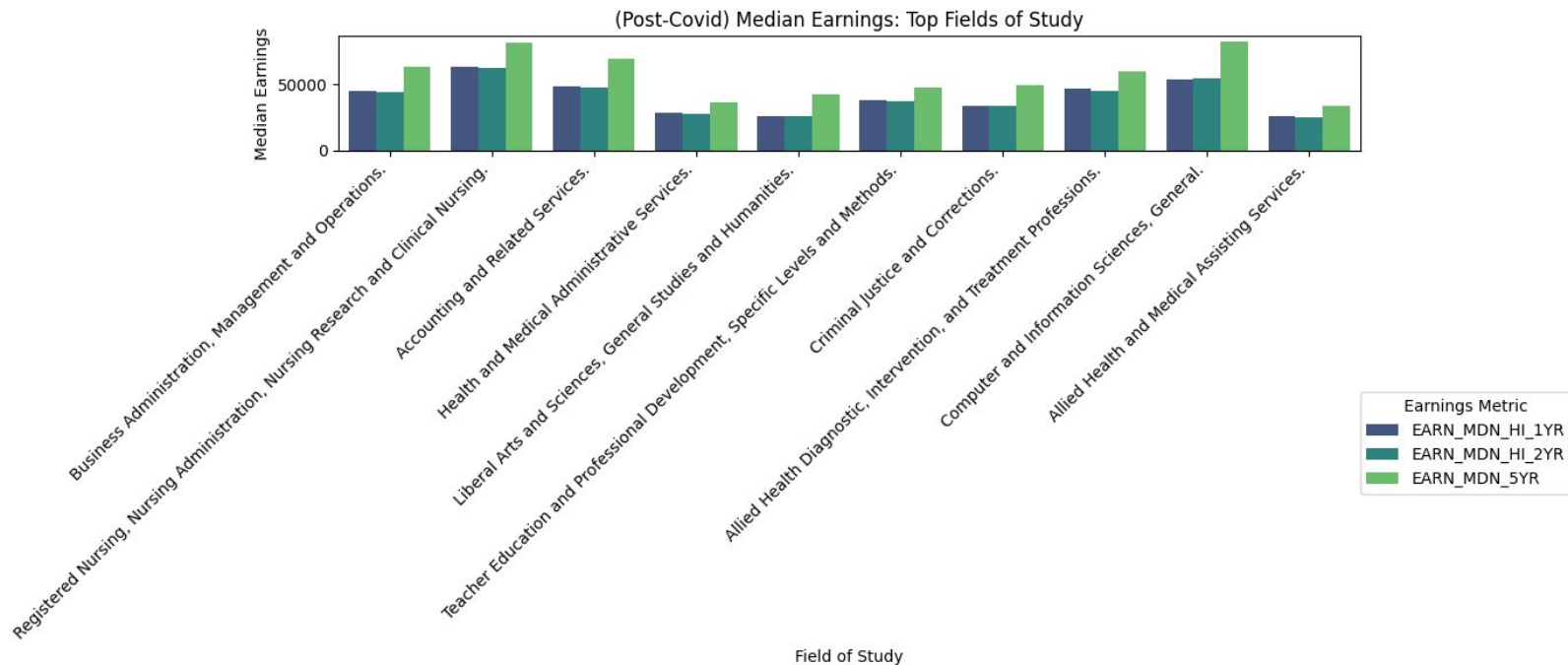
Descriptive Statistics (pre-Covid19)



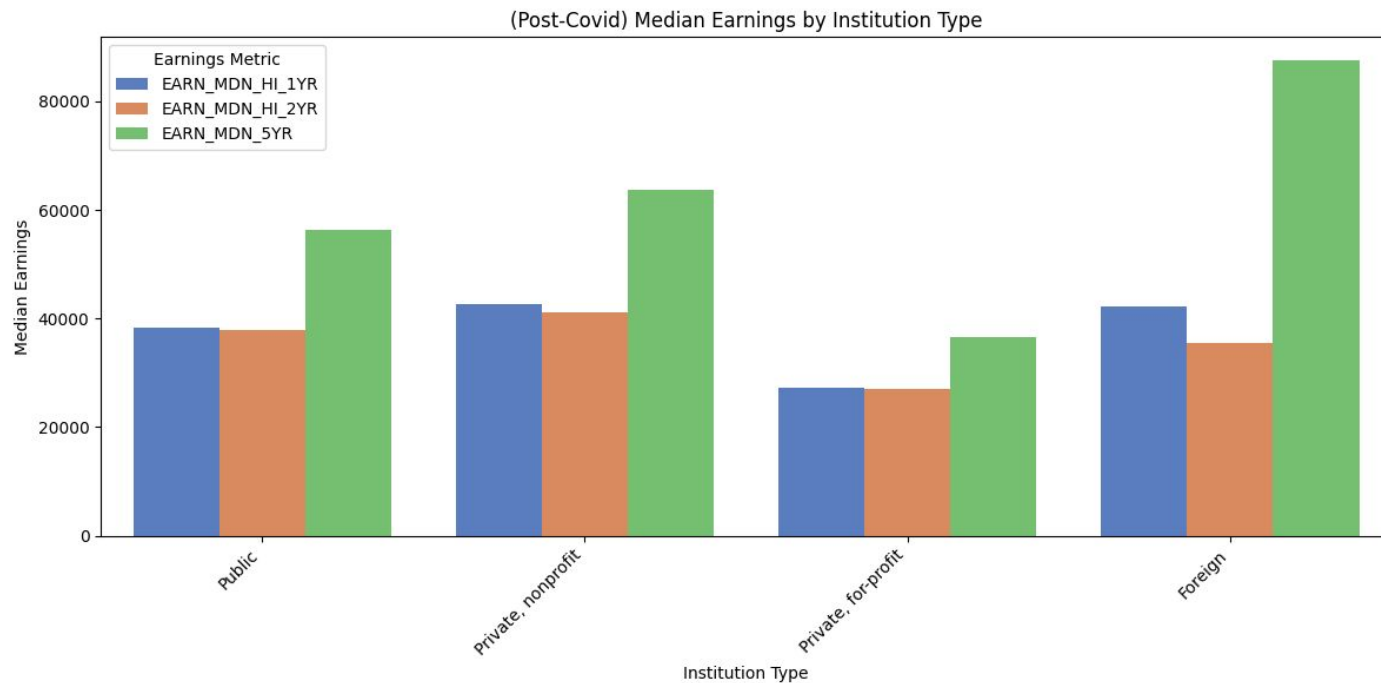
Descriptive Statistics (pre-Covid19)



Descriptive Statistics (post-Covid19)



Descriptive Statistics (post-Covid19)



K-Means Clustering

- Technique that assigns data points to one of the K clusters depending on their distance from the center of the clusters. Objective is to divide the dataset into a specific number of groups for comparison.
- Graduates were grouped according to salary (1 year - pre-pandemic, 4 & 5 years - post-pandemic), Distance, Control, CIPDESC (industry)
 - categorical variables encoded for comparison
- Data was split into 4 clusters & grouped for pre & post-pandemic

K-Means Clusters

Pre-Pandemic:

	CONTROL	DISTANCE	CIPDESC	EARN_MDN_1YR
Cluster				
0	2.735620	1.076030	258.061118	0.012225
1	1.328761	0.890837	118.465215	-0.663694
2	2.179845	2.755048	147.199094	0.337699
3	2.824635	1.057498	73.839770	0.238149

Post-Pandemic:

	CONTROL	DISTANCE	CIPDESC	EARN_MDN_4YR	EARN_MDN_5YR
Cluster					
0	2.738392	1.252352	259.535302	-0.241389	-0.238006
1	2.445670	1.296994	151.242753	1.777956	1.767886
2	2.744136	1.335528	76.267221	-0.133942	-0.125655
3	1.170662	1.177597	122.052766	-0.786107	-0.799417

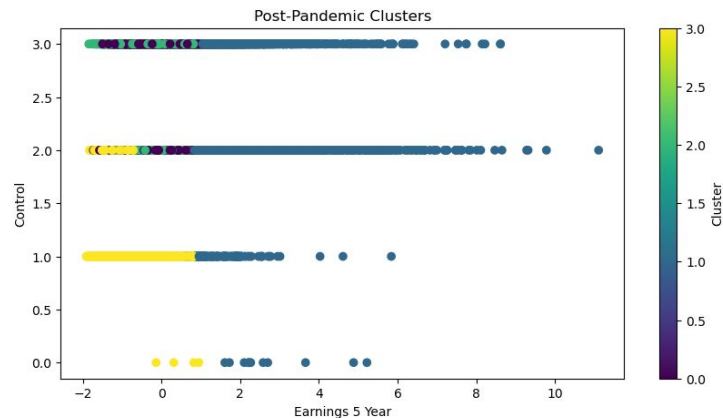
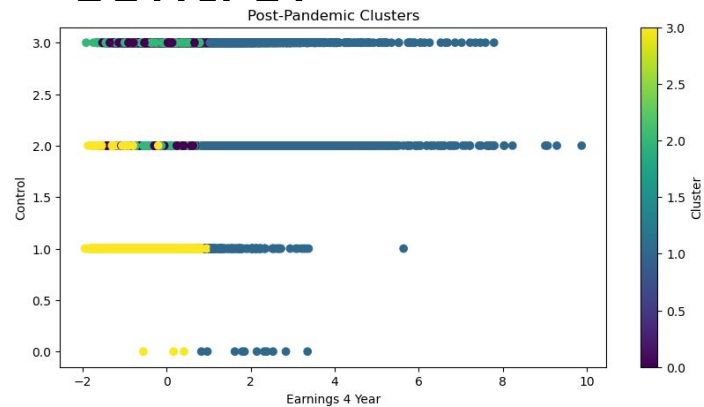
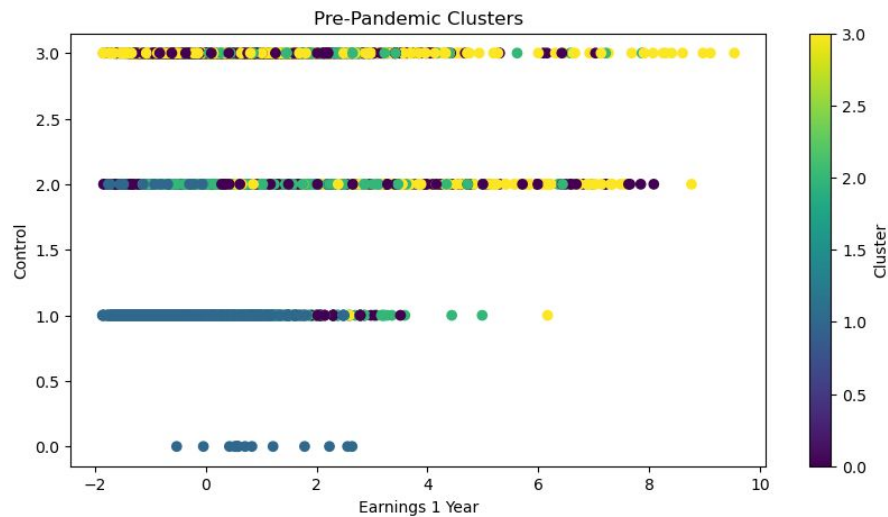
Cluster 0: Control, Distance & CIPDESC remains constant, earnings decreased post-pandemic

Cluster 1: Control, Distance & CIPDESC increased alongside earnings

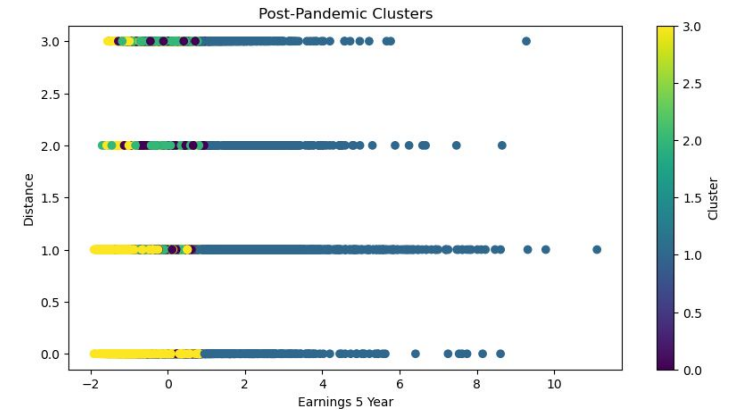
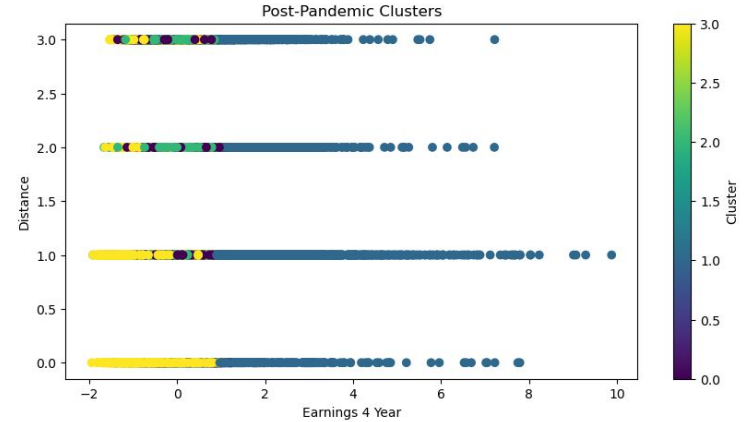
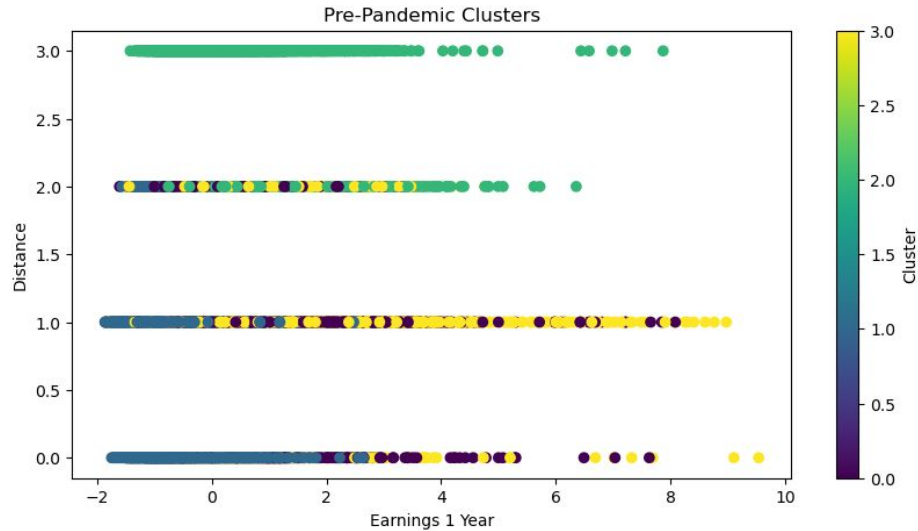
Cluster 2: Control has increased, while Distance, CIPDESC & earnings have decreased

Cluster 3: Control significantly decreased, distance & CIPDESC increased, earnings decreased significantly

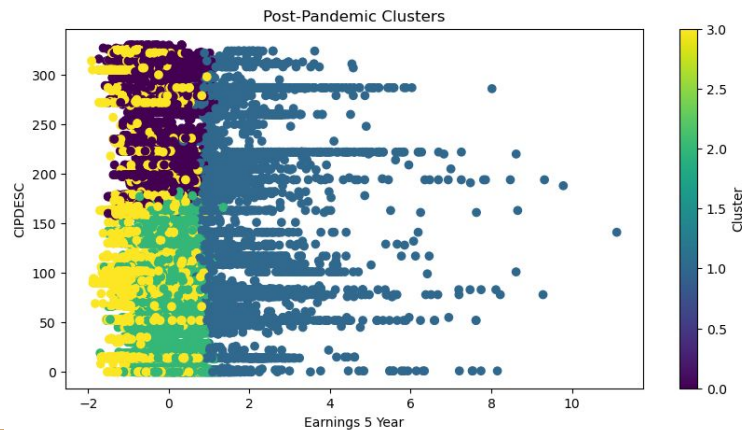
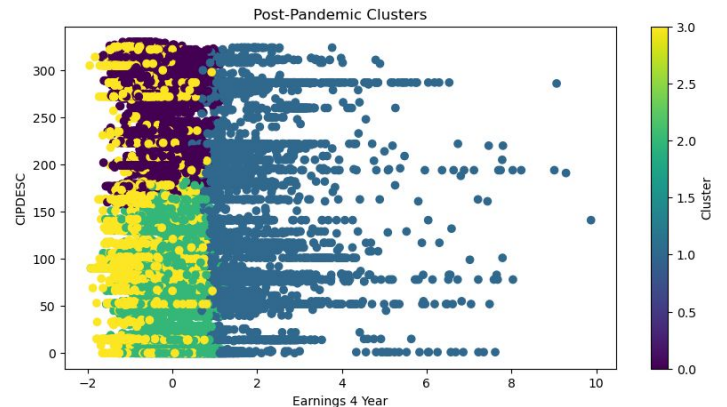
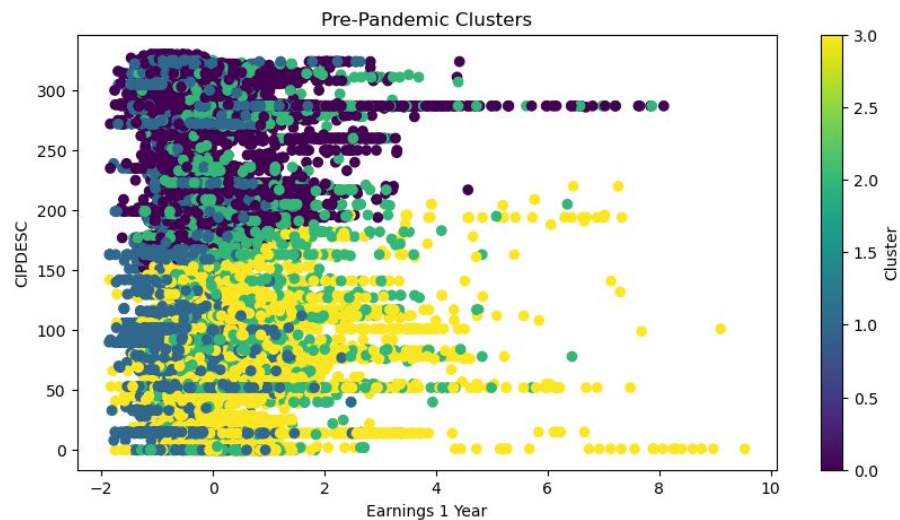
K-Means Clustering Results - Control



K-Means Clustering Results - Distance



K-Means Clustering Results - CIPDESC



Results

- Cluster 1 experienced significant improvements, especially in earnings
 - Increase in Control & Distance indicates an adaptive shift during the pandemic
- Cluster 0 showed slight decrease in earnings, but not much change
- Clusters 2 & 3 showed the most negative changes in all categories, implying they were hit the hardest during the pandemic
 - Decrease in control for cluster 3 could indicate more governmental influence post-pandemic
 - Changes in CIPDESC for both indicate socio-economic shift

Synthetic control

——Method Overview

A statistical technique for causal inference that constructs a weighted combination of control units to create a synthetic counterpart for the treated unit.

Compares outcomes of the treated unit with its synthetic control before and after the intervention to estimate the treatment effect.

Advantages:

1. Handles cases where randomization is infeasible or unethical.
2. Accounts for unobservable confounders, provided they are time-invariant.
3. Allows visualization of treatment effects over time.

XGBoost

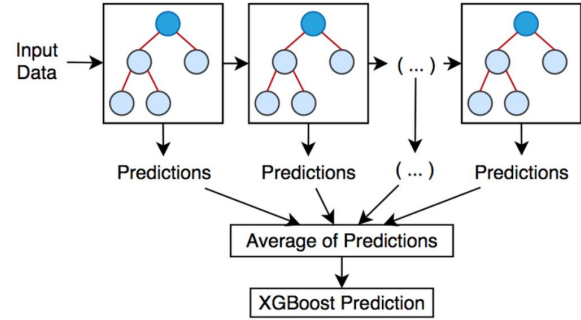
A powerful machine learning algorithm based on gradient boosting.

Why XGBoost?

High Accuracy: XGBoost has shown to deliver high predictive accuracy, especially in regression tasks like predicting salaries.

Handles Missing Data: XGBoost can handle missing values without the need for imputation, making it well-suited for real-world data

Scalability: It is capable of handling large datasets and can be easily parallelized, making it suitable for big data scenarios.

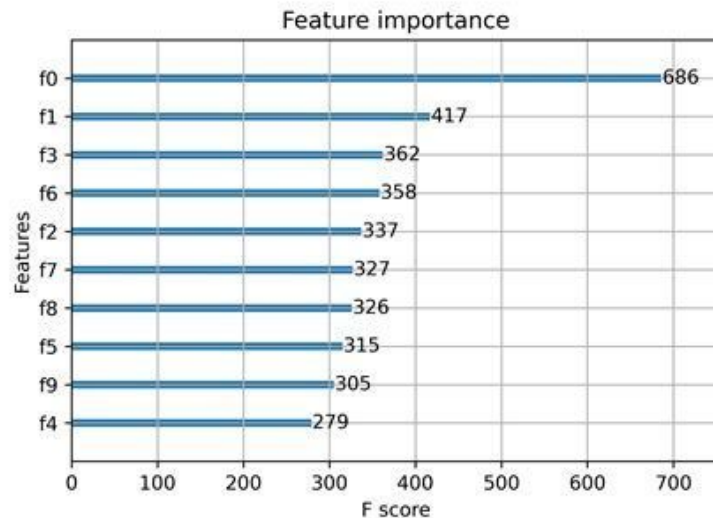


Feature Engineering

Adding variables from different perspectives: admission rate, tuition fee, average SAT, school type e.g

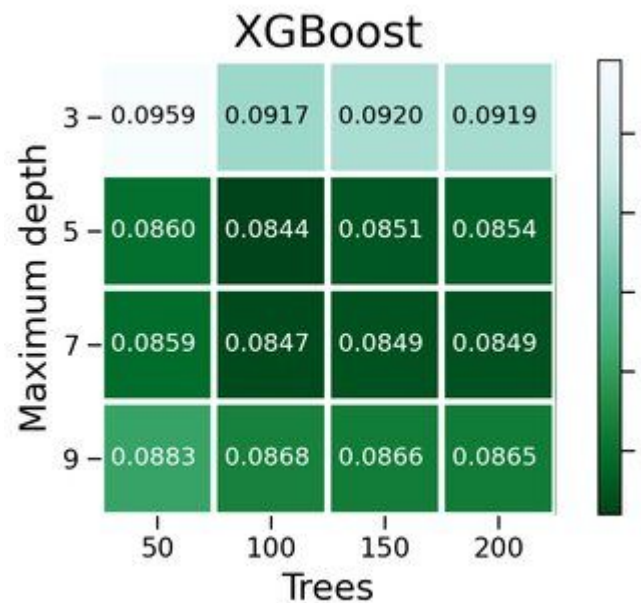
Adding and Deleting features according to feature importance

Creating new features by making calculations between available features.



Parameter Grid Search

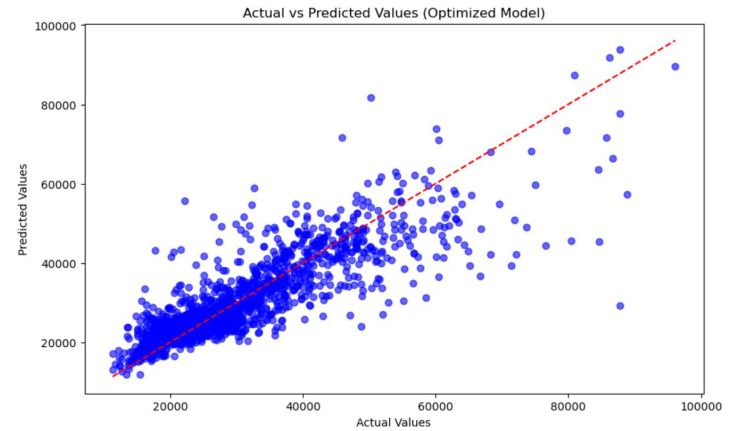
Using Parameter Grid Search to find out the optimized combination for each parameter



Model Construction

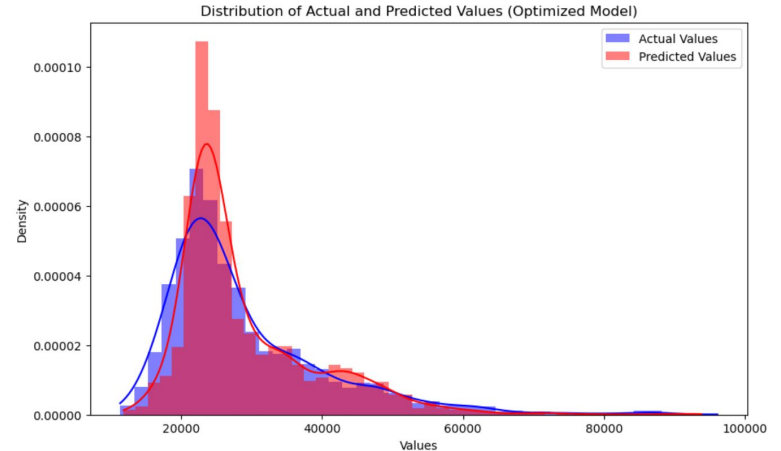
Actual vs Predicted Values (Optimized Model):

- Model predictions align well with actual values, indicating strong performance.



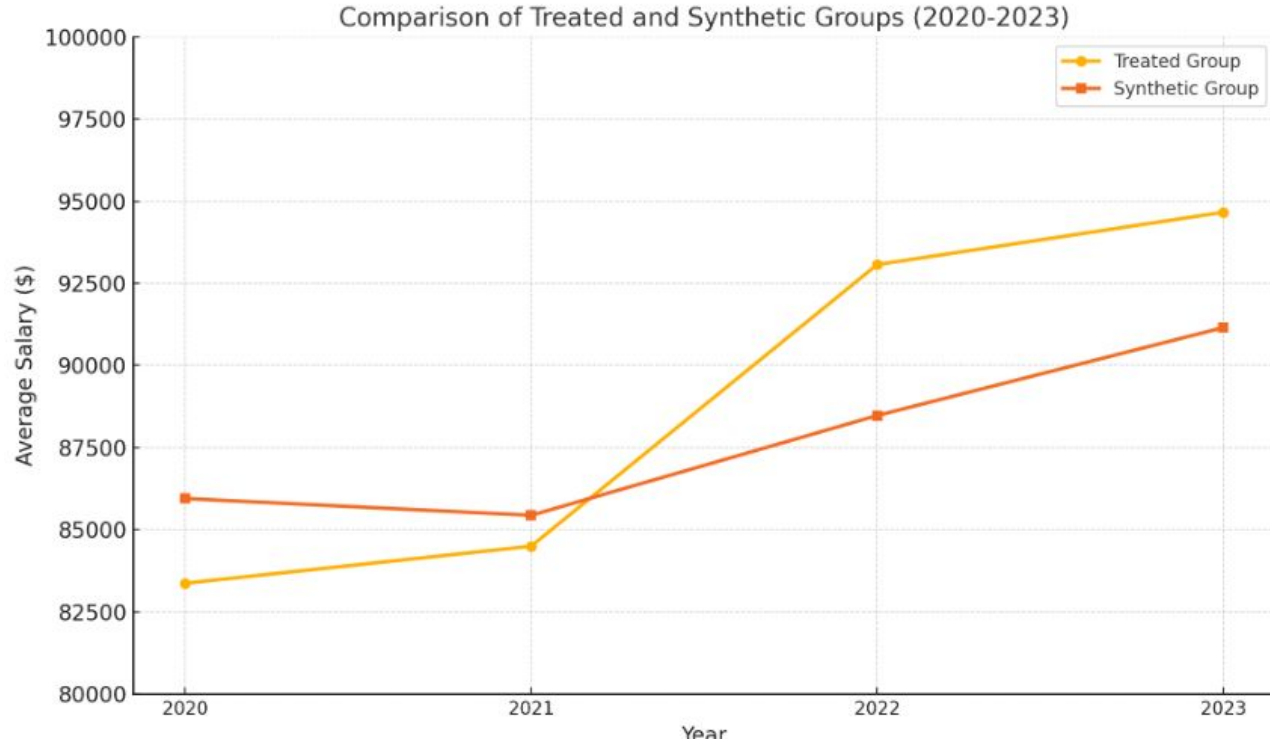
Distribution of Actual and Predicted Values (Optimized Model):

- Predicted values closely match actual values, with slight underestimation at higher values.



Finalized model reached coefficient of determination of 0.743

Treated group and Synthetic group



Result

Pandemic did had a negative impact on graduate wage at early state.

This impact was quickly recovered in the second year of the pandemic.

The impact of the pandemic on graduate salaries is not significant from an overall perspective, which is contrary to our expectations, and it cannot be proven that the pandemic caused a decline in graduate salary levels.

Linear Regression Model - Before Covid-19 (2015-2019)

OLS Regression Results

```
=====
Dep. Variable:          EARN_MDN_1YR      R-squared:                0.616
Model:                  OLS               Adj. R-squared:           0.598
Method:                 Least Squares     F-statistic:             33.56
Date:                   Thu, 05 Dec 2024  Prob (F-statistic):      0.00
Time:                   16:48:45          Log-Likelihood:          -5.8326e+05
No. Observations:      53307             AIC:                    1.171e+06
Df Residuals:          50872             BIC:                    1.193e+06
Df Model:               2434
Covariance Type:       nonrobust
```

Y variables: Median Earnings One Year After Graduation

X variables: Type of institution (e.g., Public, Private Non-Profit, Private For-Profit), City where the institution is located, State where the the Institution is located, Description of the program/course of Study

Baseline: Foreign, Artesia, AK(Alaska), Accounting and Related Services

Significant Positive Associations (2015-2019)

- **Academic Disciplines:**

- **Dentistry and Oral Sciences** (131443.6579)
- **Medical Clinical Sciences/Graduate Medical Studies** (72309.04445)
- **Computational Science** (46900.41516)
- **Security Policy and Strategy** (43822.25656)
- **Military Applied Sciences** (38859.46023)
- **Systems Engineering** (38762.37032)
- **Pharmacy, Pharmaceutical Sciences, and Administration** (38154.813299)
- **Optometry** (36329.77695)

- **Locations:**

- **Redwood City** (48069.49773)
- **Stanford** (34999.08587)
- **Los Altos Hills** (34398.8823)
- **Alpena** (32405.780952)
- **Castine** (32276.08527)
- **Addison** (31739.13614)
- **North Chicago** (31219.94885)
- **Charles Town** (25889.53322)
- **Hanover** (25221.53761)
- **Redmond** (24698.94803)
- **Wake Forest** (24658.29815)

Significant Negative Associations (2015-2019)

- **Academic Disciplines:**

- **Crafts/Craft Design, Folk Art and Artisanry** (-34227.39361)
- **Dance** (-33733.88986)
- **Human Biology** (-33019.72991)
- **Drama/Theatre Arts and Stagecraft** (-32588.68059)
- **Visual and Performing Arts, Other** (-30577.33331)
- **Literature** (-30534.25189)
- **Intercultural/Multicultural and Diversity Studies** (-29454.4787)
- **Leisure and Recreational Activities** (-29130.41826)

- **Locations:**

- **Cupertino** (-33371.92613)
- **Grundy** (-32944.86989)
- **South Royalton** (-27267.34705)
- **Yuma** (-26980.62045)
- **Sierra Vista** (-25324.22728)
- **El Centro** (-24186.4456)
- **Watsonville** (-20774.16137)
- **Charlottesville** (-18315.37149)

-----ALL-----

- **Institution Type:**

- **Private, for profit** (-13273.178)
- **Public** (-6099.821329)

Linear Regression Model - After Covid-19 (2023-2024)

OLS Regression Results

=====			
Dep. Variable:	EARN_MDN_1YR	R-squared:	0.619
Model:	OLS	Adj. R-squared:	0.600
Method:	Least Squares	F-statistic:	33.22
Date:	Tue, 03 Dec 2024	Prob (F-statistic):	0.00
Time:	04:03:18	Log-Likelihood:	-5.7411e+05
No. Observations:	52005	AIC:	1.153e+06
Df Residuals:	49580	BIC:	1.175e+06
Df Model:	2424		
Covariance Type:	nonrobust		

Y variables: Median Earnings One Year After Graduation

X variables: Type of institution (e.g., Public, Private Non-Profit, Private For-Profit), City where the institution is located, State where the the Institution is located, Description of the program/course of Study

Baseline: Foreign, Artesia, AK(Alaska), Accounting and Related Services

Significant Positive Associations (2023-2024)

- **Academic Disciplines:**

- **Dental Residency Programs** (175951.455)
- **Medical Clinical Sciences/Graduate Medical Studies** (99549.59016)
- **Optometry** (40631.18572)
- **Systems Engineering** (39,067.748573)
- **Veterinary Medicine** (38569.2151)
- **Pharmacy, Pharmaceutical Sciences, and Administration** (37391.59959)
- **Computational Science** (36822.14606)
- **Paper Science and Engineering** (32649.18761)

- **Locations:**

- **Stanford** (32364.55975)
- **Los Altos Hills** (31074.03447)
- **Castine** (29268.6876)
- **Versailles** (28191.20425)
- **North Chicago** (27129.83384)
- **Hanover** (26443.88236)
- **Yankton** (25184.05019)
- **DC** (23875.03202)
- **Charles Town** (21748.89292)
- **Evanston** (21189.24552)

Significant Negative Associations (2023-2024)

- **Academic Disciplines:**

- **Energy and Biologically Based Therapies** (-48125.69703)
- **Crafts/Craft Design, Folk Art and Artisanry** (-37446.51395)
- **Literature** (-36490.69432)
- **Drama/Theatre Arts and Stagecraft** (-36202.40051)
- **Dance** (-36072.32928)
- **Human Biology** (-35315.95924)
- **Cultural Studies/Critical Theory and Analysis** (-34393.10584)
- **Visual and Performing Arts, Other** (-33907.90569)

- **Locations:**

- **Miami Lakes** (-54132.23212)
- **Pablo** (-45386.06377)
- **Grundy** (-44737.03914)
- **Alhambra** (-39865.63243)
- **Yuma** (-39085.63969)
- **South Royalton** (-38762.47732)
- **Cuthbert** (-36007.6653)
- **El Centro** (-34781.89038)
- **Holbrook** (-33423.61423)
- **Sierra Vista** (-30977.42576)

- **Institution Type:**

- **Private, for-profit** (-16817.09554)
- **Public** (-8848.447948)

Results

Variation Across Disciplines:

- Strong positive associations (compared to Accounting and Related Services)
 - **Healthcare and Medical Fields, Engineering Disciplines, Mathematics, Computer Science, Operations Research.**
- **Strong negative associations** (compared to Accounting and Related Services)
 - **Creative and Performing Arts, Humanities and Social Sciences, Niche Fields** (e.g. Therapeutic studies, Outdoor Education, Precision Production): strong negative associations

Institution-Type Patterns:

- On average, graduates **from Private (For-Profit) and Public Institutions** earn less compared to those from foreign institutions. **Private (for-profit)** has a larger negative coefficient, suggesting a bigger salary gap with foreign graduates compared to public institutions.

City-Level Patterns:

- **Shift in Negative Association: Post-COVID**, cities in economically disadvantaged areas (e.g., Miami Lakes, Pablo, Alhambra, Cuthbert) now show significant negative associations with graduate salaries, underperforming Artesia. **Pre-COVID**, the extreme negative impact on salaries was primarily associated with certain program disciplines. Post-COVID, the focus of extreme negative associations has shifted towards specific locations, indicating the growing importance of institutional location in salary outcomes.
- Strong positive associations with graduate salaries suggest locations like **Stanford, Los Altos Hills, and Castine** are linked to prestigious institutions, medical/research hubs, or major industry clusters, contributing to higher salary outcomes compared to Artesia.
- Strong negative associations with salary outcomes suggest areas like **Grundy, South Royalton, and Yuma** may lack academic or professional opportunities comparable to Artesia. Smaller towns or rural areas with more limited program specializations or a focus on local/regional needs might not translate into competitive salary outcomes.

Summary

- The impact of pandemic on college graduate job market on the impact of the pandemic on graduate salaries is not significant from an overall perspective
- Although COVID-19 pandemic reduced salaries for some groups & industries, others were able to adapt to the pandemic and experience growth through changes in geographic location & sector of work

References

Szczygielski, Jan Jakub, et al. "The Impact and Role of Covid-19 Uncertainty: A Global Industry Analysis." *International Review of Financial Analysis*, North-Holland, 2 July 2021, www.sciencedirect.com/science/article/pii/S1057521921001708 [accessed Nov. 10, 2024].

O. Khlystova, Y. Kalyuzhnova, and M. Belitski, "The impact of the COVID-19 pandemic on the Creative Industries: A Literature Review and Future Research Agenda," *Journal of Business Research*, <https://www.sciencedirect.com/science/article/pii/S0148296321007141> [accessed Nov. 10, 2024].